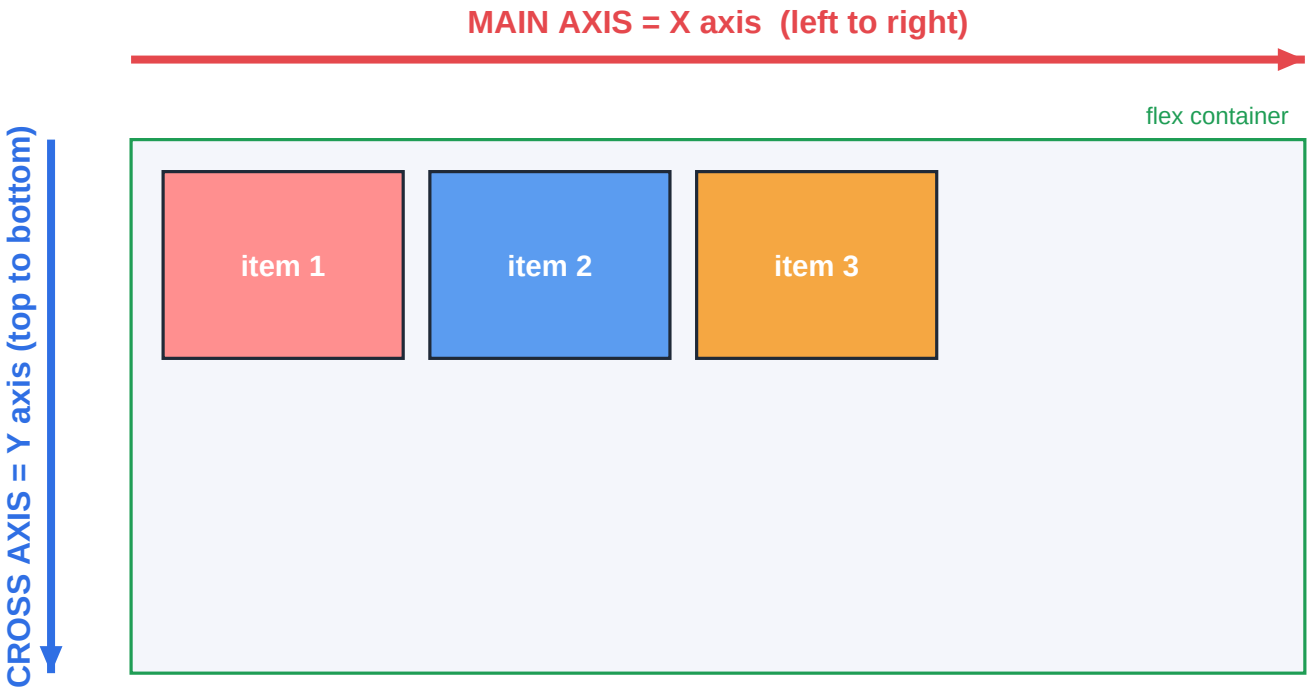
Flexbox: Main Axis and Cross Axis

Page 1 - flex-direction: row (the DEFAULT)

Step 1: Write this on the parent (container):

```
.container { display: flex; }
```

The parent = FLEX CONTAINER. Its direct children = FLEX ITEMS.



In ROW direction:

Main axis	= X axis, horizontal (left to right)
Cross axis	= Y axis, vertical (top to bottom)
Items flow	= side by side, in a row
justify-content	= moves items along MAIN axis (left / right)
align-items	= moves items along CROSS axis (up / down)

Easy way to remember

MAIN axis = the direction your items are lined up.

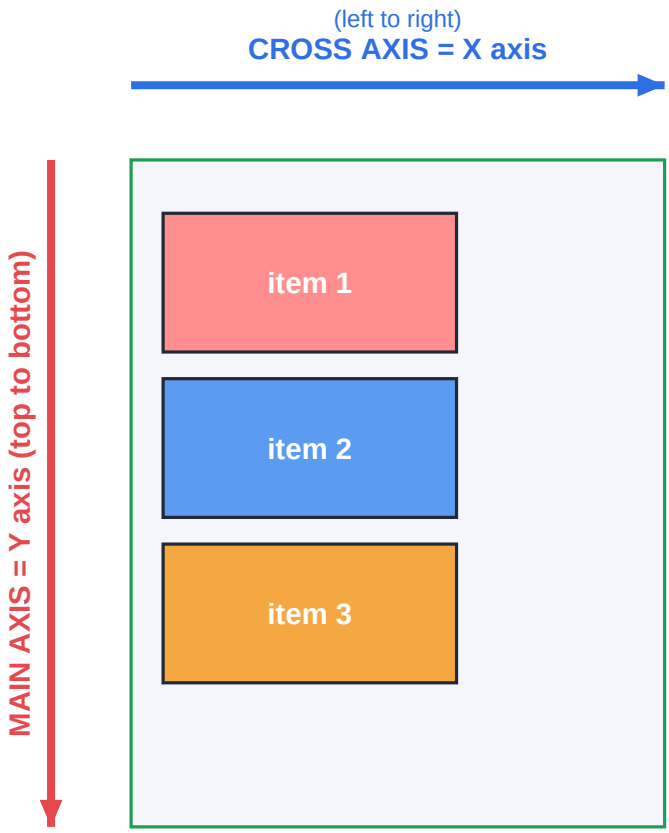
CROSS axis = the other direction (always at 90 degrees to main).

justify-content -> MAIN. align-items -> CROSS.

What happens when you flip to column?

Page 2 - flex-direction: column (the axes SWAP)

```
.container { display: flex; flex-direction: column; }
```



Before and after the flip

	ROW	COLUMN
Main axis	X (horizontal)	Y (vertical)
Cross axis	Y (vertical)	X (horizontal)
Items flow	left to right	top to bottom
justify-content moves items	left / right	up / down
align-items moves items	up / down	left / right

The BIG idea

Axes do NOT change their names. They change their DIRECTION.

```
flex-direction: row     ->  main = X (horizontal),  cross = Y (vertical)
flex-direction: column  ->  main = Y (vertical),    cross = X (horizontal)
```

So justify-content and align-items swap jobs on screen, but still
justify = MAIN axis, align = CROSS axis.

Other directions

`row-reverse`	main axis still X, but items start from the RIGHT
`column-reverse`	main axis still Y, but items start from the BOTTOM

Tip: for column + justify-content to work, the container needs a height (example: height: 100vh;). Otherwise it has no extra space to spread items.

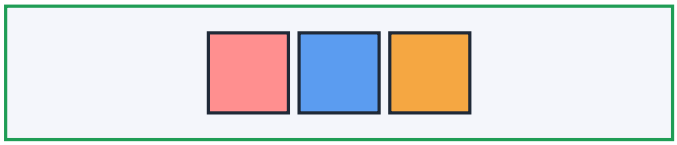
All values: justify-content and align-items

Page 3 - shown for flex-direction: row

justify-content (MAIN axis, left/right in row)

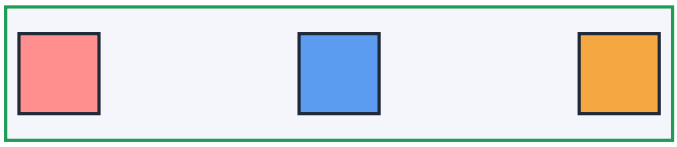
`flex-start`

`center`

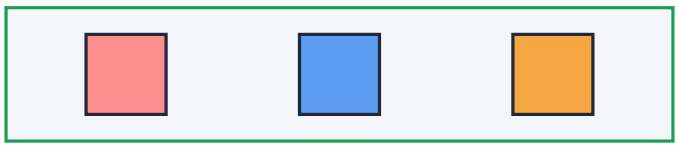


`flex-end`

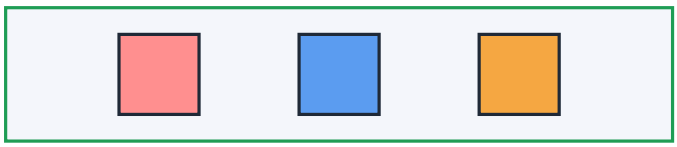
`space-between`



`space-around`

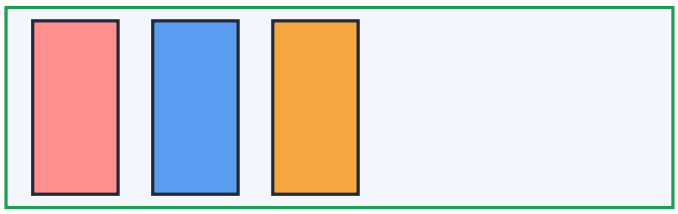


`space-evenly`

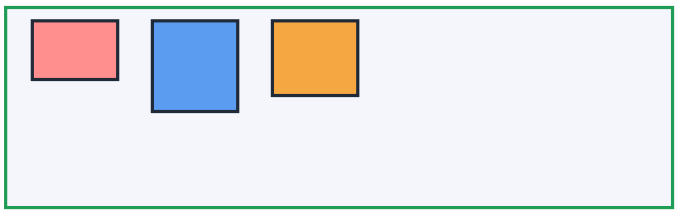


align-items (CROSS axis, up/down in row)

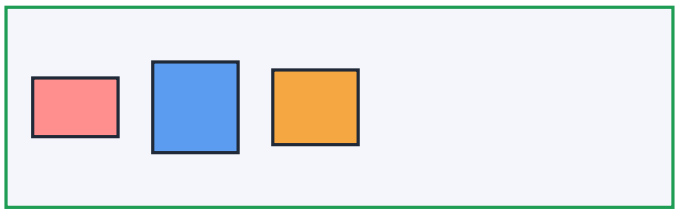
`stretch`



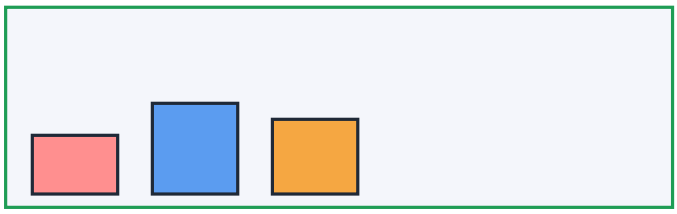
`flex-start`



`center`



`flex-end`



Other handy properties

`gap: 10px;`	space between items
`flex-wrap: wrap;`	items go to next line when no space
`align-self: center;`	on ONE item: overrides align-items
`flex-grow: 1;`	on an item: takes the leftover space
`order: 2;`	change the position of an item
`align-content: center;`	spreads lines when flex-wrap is used